



Research paper

Molecular modelling of active oil droplet propulsion: Insights from dissipative particle dynamics simulation

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HIGHLIGHTS

- DPD simulations reproduced self-propelled motion of oil droplets at the molecular level.
- Larger oil droplet radii resulted in proportional increases in propulsion velocity.
- A 2% interfacial tension difference significantly increased propulsion speed.
- Surfactant structure impacted oil–oil interactions, influencing propulsion velocity.
- The DPD approach explained molecular mechanisms in non-equilibrium active systems.

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ABSTRACT

This study employed dissipative particle dynamics (DPD) simulations to investigate the self-propelled motion of oil droplets in water–oil–surfactant systems. It is the first attempt to replicate self-propulsion models of oil droplets at the molecular level, contrasting previous simulations focused on Brownian motion and hydrodynamic behaviour of colloidal particles. The DPD model reproduced droplet propulsion and visualised internal Marangoni flow, showing that larger droplet radii and greater interfacial tension differences increase propulsion speeds. Additionally, surfactants with stronger oil–oil repulsion enhanced propulsion speed, suggesting that surfactant-induced local structures are crucial for the self-propulsion mechanism.

1. Introduction

The concept of ‘active matter’, which refers to the collective motion of self-propelled particles in nonequilibrium systems [1,2], has ignited robust discussions spanning various scales. These discussions range from macroscopic assemblies, such as flocks of birds or schools of fish, to microscale occurrences, such as bacterial swimming [3–5]. The self-propulsion of microscale swimmers requires breaking the isotropic symmetry between the particles and their environment, either spontaneously or driven by external stimuli. Microswimmers fall into two categories. Solid microswimmers, such as Janus or colloidal particles, exhibit inherent breaking of isotropic symmetry owing to their geometric shapes [6,7]. Meanwhile, swimmers feature isotropic particles, but their symmetry is disrupted by interactions with the surrounding fluid [8–10]. The latter has gained significant attention owing to its design simplicity and high responsiveness to chemical stimuli, making it promising for biomedical and engineering applications [11–15].

Microscopic liquid droplets in water/organic systems can autonomously propel themselves into active droplets in a nonequilibrium state. This phenomenon arises from surface reactions, such as solubilisation at interfaces and interfacial flows generated by solute gradients. The self-propulsion of oil droplets in surfactant systems is an example of such microscale active droplets. This phenomenon is theoretically attributed to the Marangoni flow owing to the surface tension gradient on the oil droplet interface, subsequently inducing internal flow of the droplet [16]. Marangoni flow occurs when there is a variation in the surface tension at the interface of a liquid system owing to temperature or concentration gradients, leading to a flow driven by shear stresses induced by uneven surface tension. The coffee-ring effect is a well-known example of this phenomenon [17].

Banno et al. developed multiple models to elucidate this phenomenon and identified the driving mechanism originating from the surface tension gradient-induced Marangoni flow [18–20]. These oil

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droplets have garnered attention as energy-efficient material transport systems that do not require external forces. However, these studies have predominantly adopted a continuum (fluid) perspective. Understanding the chemical interactions between these elements is indispensable for understanding the essence of oil droplet propulsion in water/oil/surfactant systems.

While simulations have focused on Brownian motion in microswimmers [21] and the hydrodynamic behaviour of colloidal particles [22], there have been no molecular-level simulations replicating self-propulsion models of oil droplets in surfactant systems. To the best of our knowledge, this is the first study to replicate this phenomenon using molecular simulations. Additionally, by employing models that mimic the self-propulsion of oil droplets, we aimed to theoretically explore how interactions among water, oil, and surfactants influence oil droplet propulsion.

2. Model and methods

2.1. Dissipative particle dynamics

This study used the dissipative particle dynamics (DPD) method [23–25], a coarse-grained molecular simulation technique, to reproduce the self-propulsion of oil droplets in water–oil–surfactant systems. Although the self-propulsion of oil droplets studied in experimental systems is thought to be achieved by chemical interactions between molecules, it is challenging to measure these interactions in nonequilibrium phenomena. Although direct observation by molecular simulation has the potential to significantly advance our understanding of this phenomenon, the experimentally observed spatiotemporal scales (on the order of $10^{-1}\sim 10^1$ μm and $10^1\sim 10^3$ s) cannot be reproduced by classical molecular dynamics (MD) simulations. Therefore, the DPD method, which incorporates a coarse-graining technique, can handle substantial spatiotemporal scales compared to MD (although smaller than those in experiments). In addition, because it is possible to focus only on pure interactions, it is expected to capture the essence of self-propelling phenomena. The DPD method has been used for various types of motions in the past [26–31] and is the most suitable tool for reproducing the present system.

The particle motion in DPD is described by Newton's equations of motion. The force acting on each particle (i) is the sum of three terms.

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i = \sum_{j \neq i} \mathbf{F}_{ij}^C + \sum_{j \neq i} \mathbf{F}_{ij}^D + \sum_{j \neq i} \mathbf{F}_{ij}^R. \quad (1)$$

Here, $\mathbf{F}_{ij}$ is the force exerted on particle i by particle j , and $\mathbf{F}_{ij}^C$, $\mathbf{F}_{ij}^D$, and $\mathbf{F}_{ij}^R$ are conservative, dissipative, and random forces, respectively. The expressions for the conservative force $\mathbf{F}_{ij}^C$, dissipative force $\mathbf{F}_{ij}^D$, and random force $\mathbf{F}_{ij}^R$ in DPD are as follows:

$$\mathbf{F}_{ij}^C = a_{ij} \left(1 - \frac{|\mathbf{r}_{ij}|}{r_c} \right) \mathbf{n}_{ij}, \quad |\mathbf{r}_{ij}| \leq r_c \quad (2)$$

$$\mathbf{F}_{ij}^D = -\gamma \omega^D(|\mathbf{r}_{ij}|) (\mathbf{n}_{ij} \cdot \mathbf{v}_{ij}) \mathbf{n}_{ij}, \quad |\mathbf{r}_{ij}| < r_c \quad (3)$$

$$\mathbf{F}_{ij}^R = \sigma \omega^R(|\mathbf{r}_{ij}|) \xi_{ij} \Delta t^{-1/2} \mathbf{n}_{ij}, \quad |\mathbf{r}_{ij}| \leq r_c \quad (4)$$

Here, a_{ij} is the repulsion parameter between particles i and j , which serves as the sole parameter providing substance-specific information to the coarse-grained particles in DPD. $\mathbf{r}_{ij}$, $\mathbf{v}_{ij}$, and $\mathbf{n}_{ij}$ are the position, velocity, and unit vectors pointing from particle i to particle j , respectively. r_c is the cutoff distance beyond which the interactions between the dissipative particles are zero. ω^D and ω^R are the weighting functions, and γ is the friction parameter. The noise parameter σ is related to the temperature T and Boltzmann constant k_B as $\sigma^2 = 2\gamma k_B T$.

Generally, reduced units are used to report DPD results. The DPD unit of length is the cutoff radius r_c , the unit of mass is the bead mass m , and the unit of energy is $k_B T$. Therefore, the time unit is defined as $\tau = \frac{\sqrt{m r_c}}{k_B T}$.

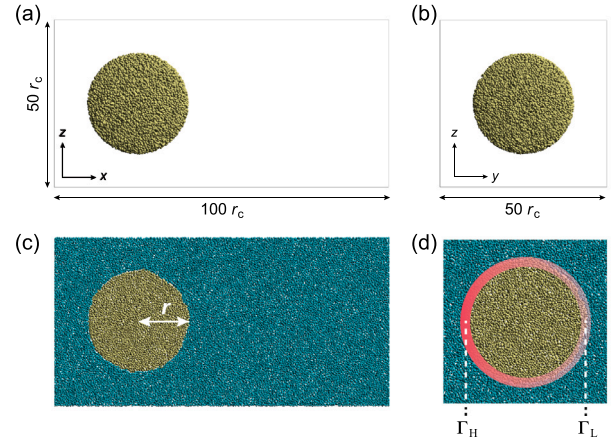


Fig. 1. Simulation system in the (a) x-z plane and the (b) y-z plane. For clarity, only oil droplets are depicted, while water particles are omitted from the illustration. (c) Cross-sectional view of the initial configuration. (d) To reproduce the surface tension Γ difference in this model, the chemical interaction between water and oil changes gradually in the droplet's x direction forward and backward.

2.2. Simulation model and conditions

In an oil-droplet propulsion system, the propulsion phenomenon of oil droplets is caused by the interaction of water, oil, and surfactant molecules. However, this study focused only on the explicit treatment of water and oil molecules. In contrast, surfactant molecules are treated implicitly as having the effect of lowering the surface tension at the interface. This study aimed to obtain fundamental knowledge about this propulsion phenomenon. Fig. 1(a), (b), and (c) show the initial configurations of the simulated systems. The system consisted of 750,000 particles, including 12,428–100,264 oil particles, depending on the oil drop radius r .

The water and oil particles are labelled by the letters 'w' and 'o', respectively. The interaction parameters a_{ij} between the same kinds of particles were set to $a_{ww} = a_{oo} = 25 k_B T$. On the other hand, the magnitude of the interaction (a_{wo}) between water and oil particles varies under the influence of the surfactant concentration at the interface. In other words, a_{wo} gradually increases from forward to backward concerning the direction of droplet propulsion to reproduce the difference in surface tension Γ due to surfactant concentration, as shown in Fig. 1(d). Here, Γ_H corresponds to the a_{wo} value at the end of the droplet with the lowest surfactant concentration and Γ_L to the a_{wo} value at the front of the droplet with the highest surfactant concentration; in the most typical system, their values, Γ_H and Γ_L are $100 k_B T$ and $50 k_B T$, respectively. The particle with the largest position for the direction of droplet propulsion is assigned the smallest a_{wo} value (Γ_L), while the particle with the smallest position is assigned the largest a_{wo} value (Γ_H) from moment to moment. Other oil particles are given a_{wo} to interpolate linearly concerning their positions.

All simulations were performed in a constant-volume and constant-temperature ensemble, and periodic boundary conditions were applied in all three dimensions. The volume of the simulation box is $100 \times 50 \times 50 r_c^3$. The noise parameter σ , friction parameter γ , and time step Δt were set as 3.0, 4.5, and 0.01 τ , respectively. The system temperature was set to $0.5 k_B T$ to minimise the influence of thermal fluctuations.

3. Results and discussions

3.1. Reproduction of the self-propelled oil droplet motion

Initially, we established the system depicted in Fig. 1 and endeavoured to replicate the self-propagation of oil droplets through molecular

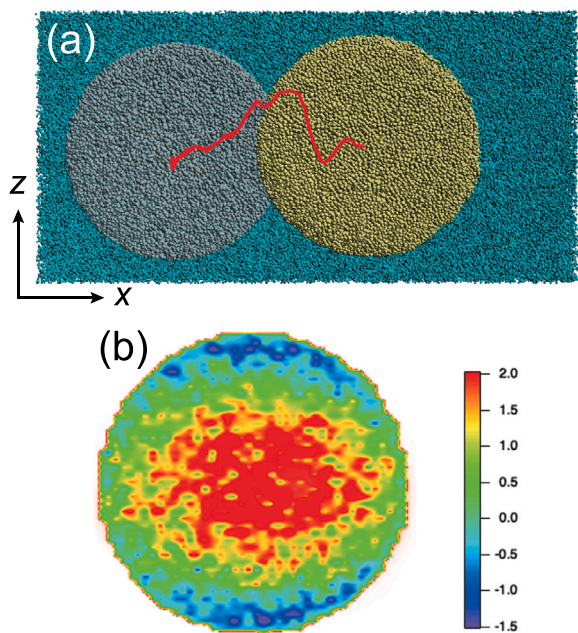


Fig. 2. (a) Overlaid snapshots at initial state (grey droplet) and $t = 1000 \tau$ (yellow droplet). The red line represents the trajectory of the oil droplet centre of mass during the 1000τ . (b) Contour colour map of the velocity of oil particles constituting a droplet under steady state ($r = 20 r_c$). Here, the velocity is the propulsion velocity of the oil droplet, so it is the component in the x direction (positive in the left-to-right direction in the figure) in the fixed centre-of-mass coordinates of the oil droplet. The unit of the colour bar is r_c/τ .

simulation. The radius of the oil droplet r was designated as $20 r_c$, whereas Γ_H and Γ_L were, respectively, assigned values of $100 k_B T$ and $50 k_B T$.

After a brief interval, the oil droplet began to move gradually in the positive x direction, ultimately attaining a steady velocity. The trajectories of the centre-of-mass coordinates of the oil droplet spanning from the onset of the simulation to 1000τ are shown in Fig. 2a. The grey droplet signifies the initial state ($t = 0$), whereas the yellow droplet represents a state beyond 1000τ . The red lines indicate the trajectory. Notably, the droplet traverses in the direction of the surface tension gradient (x direction) as well as in orthogonal directions. This behaviour stems from the diminutive size of the oil droplets relative to practical experimentation, coupled with the substantial influence of thermal fluctuations. Nevertheless, even under such circumstances, the bias in the x direction is discernible. Considering the coarse-grained nature of the simulation and subsequent rescaling to physical units [25], the radius of the oil droplet in this instance approximates 12–18 nm. This implies that variations in the surface tension can exert a propulsive force even across spatial scales that are two to three orders of magnitude smaller than those observed experimentally.

Fig. 2b offers a visualisation of the internal flow within an oil droplet under steady-state conditions. Here, the oil particles constituting the droplet transform the droplet's centre-of-mass coordinate system, showcasing the time-averaged velocity of each oil particle in the x direction as a contour image map. Warm colours denote a positive velocity in the x direction, whereas cold colours indicate a velocity opposite the x direction. To enhance clarity, the contour image map persists as an x - z cross-sectional view proximate to the centre of mass. This illustrates that the inner droplet particles exhibit a positive velocity, whereas those near the interface demonstrate a negative velocity. In essence, this visualisation elucidates the manifestation of the ideal Marangoni convection arising from disparities in surface tension. These findings substantiate the reproducibility of the self-propelled motion of oil droplets through molecular simulations.

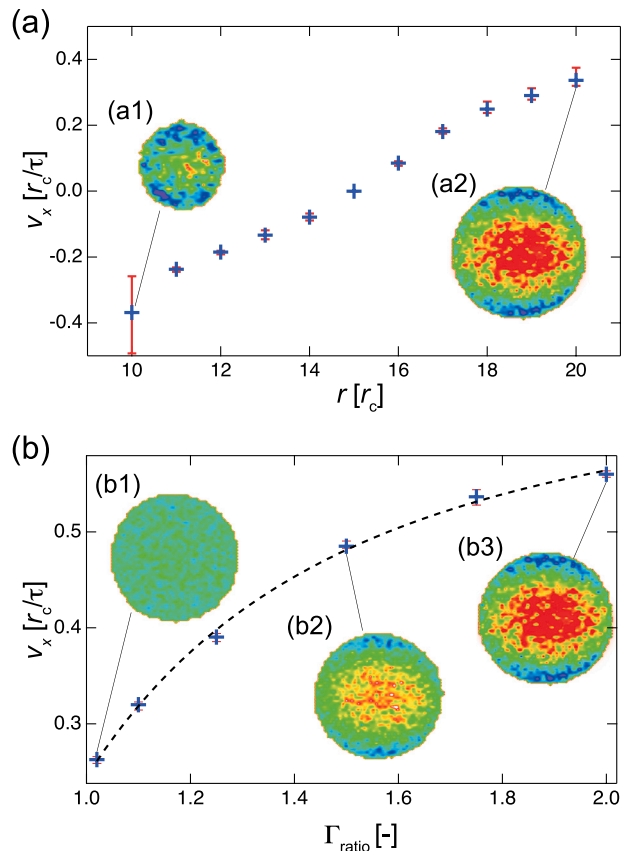


Fig. 3. Oil droplet propulsion velocity (v_x) in the x -direction dependence on (a) oil droplet radius, r and (b) surface tension difference, Γ_{ratio} . Each point is plotted based on averaged data from four independent simulations. The dashed line is the fitting curve with the equation $y = c_1 x^p + c_2$, where the constants c_1 , c_2 , and p are -0.042 , 0.067 , and -2 , respectively. Inset figures are the representative contour colour maps of the velocity of oil particles for (a1) $r = 10 r_c$, (a2) $r = 20 r_c$, (b1) $\Gamma_{\text{ratio}} = 1.02$, (b2) $\Gamma_{\text{ratio}} = 1.50$, and (b3) $\Gamma_{\text{ratio}} = 2.0$, respectively. The range of the colour bar and the positive direction of the x -direction are the same as in Fig. 2.

3.2. Effect of the oil droplet radius

An examination of the contour colour image map presented in Fig. 2b revealed that the internal flow in the positive direction (indicated by red and orange areas) exceeds the interfacial flow in the negative direction (denoted by blue and light blue areas). Consequently, one would expect the entire oil droplet to move positively because of this proportional disparity. If this assumption holds true, changes in the droplet radius should affect the propulsion velocity as the ratio of the surface area ($\propto r^2$) to volume ($\propto r^3$).

To validate the above hypothesis, we replicated the propulsive behaviour of the oil droplets while varying the droplet size from 10 to $20 r_c$ in increments of $1 r_c$ in the first series of simulations. The resulting propulsive velocities of the droplets are presented in Fig. 3a. Each data point represents the average of four simulations conducted with distinct initial conditions.

The results indicate a nearly proportional relationship between the radius of the oil droplet r and its propulsive velocity v_x , with larger radii corresponding to higher propulsion speeds. This observation aligns with the findings of the experiments conducted in a surfactant/water/oil system [32], corroborating the validity of our simulation. The behaviour observed when the droplet radius is small is of particular interest. Under the conditions of this simulation, propulsive motion was noted in the direction where the surface tension increased (negative x direction) for $r = 14 r_c$ or less. This implied a substantial area of negative velocity within the oil-droplet-driven system. Although

10 to 20 r_c droplet sizes were considered in this study, negative propulsive phenomena are expected to be observed even for smaller droplet sizes, following a balance between volume and surface area. On the other hand, the effect of thermal fluctuations eventually disappears at larger droplet radii, and the randomness shown in Fig. 2a becomes very weak. As a result, a more stable driving force is expected to be observed.

In the Marangoni convection of such a system, the interfacial flow area (interfacial thickness) remains relatively constant across the varying radii. However, the internal flow area changes as the radius varies, leading to fluctuations in the driving velocity of the oil droplets. This dynamic underscores the delicate balance between these two factors.

Although oil droplets of a petite size ($r < 10$ nm) have seldom been observed in practical experiments, their stable production could open avenues for transport in directions contrary to the typical case.

3.3. Effect of surface tension difference

In the second series of simulations, the influence of the magnitude of the surface tension difference on the oil droplet propulsion phenomenon was investigated. In this study, surfactant molecules are considered only as interactions and not explicitly treated. By maintaining the droplet radius fixed at 20 r_c , we varied the maximum surface tension (Γ_H) from 51 to 100 $k_B T$. The magnitude of the surface tension difference is denoted by $\Gamma_{\text{ratio}} = \Gamma_H / \Gamma_L$. The simulation results are shown in Fig. 3b. Velocity v_x increases with Γ_{ratio} . An examination of the internal flow diagrams (Fig. 3b1–b3) reveals a more pronounced Marangoni flow velocity among oil particles near the interface with an increasing surface tension difference. This enhanced flow near the interface leads to an excellent supply of internal flow, which serves as the primary driving force, consequently resulting in an increased oil droplet velocity.

The rate of increase in the oil droplet velocity with increasing surface tension difference was slower than that of the droplet radius r , and v_x exhibited a proportional relationship with Γ_{ratio} to a power of approximately -2 . Interestingly, the driving velocity is not to the 0.5 power of the surface tension difference but to the -2 power. According to Stokes' law, the terminal velocity when the driving force and viscous resistance are balanced is proportional to the square root of the driving force. However, the relationship between v_x and Γ_{ratio} is more appropriately fitted to the -2 power than the 0.5 power. When the surface tension is high, the energy required for the constituent oil particles to move across the droplet surface increases as soon as the droplet is significantly deformed, and conversely, the propulsion velocity decreases. As a result, the propulsion velocity is inversely proportional to the square of the surface tension difference. This suggests that the resistive force due to surface tension is nonlinear and increases quite rapidly.

Remarkably, despite the minimal surface tension difference of 2% ($\Gamma_{\text{ratio}} = 1.02$), self-propulsion was observed, albeit slower. This observation implies that in systems with large droplet sizes, such as those in the experiment, the influence of thermal fluctuations is nearly negligible, allowing for stable propulsive force generation even with slight variations in surface tension.

3.4. Effect of surfactant type

Our simulation model, which has demonstrated reliability, is well-suited for understanding oil droplet propulsion phenomena at the molecular level. This was evidenced by the successful replication of several crucial experimental outcomes, thereby instilling confidence in their accuracy and validity. Our investigation into the influence of surfactant type revealed a fascinating complexity. Previous experiments have shown variations in the self-propelled velocity based on the surfactant type, with some cases showing no propulsion phenomenon [19, 33]. Although it might be expected that a higher surface tension

difference would lead to faster droplet self-propulsion, experiments combined with surface tension measurements have demonstrated that surfactants with lower surface tension do not necessarily lead to faster self-propelled speeds [34]. This intriguing discrepancy suggests a divergence between the surfactant effects in static measurements and the nonequilibrium Marangoni flows, although the exact mechanisms remain unclear.

To investigate the effect of the surfactant type on droplet propulsion, we conducted molecular simulations focusing on the interactions between oil particles near the interface. The initial all-atom MD simulations constructed a coarse-grained model suitable for our investigation. In a water–oil–surfactant system, the surfactants primarily affect the oil droplet surface by adsorbing at the water–oil interface and altering the local structure. We examined the variation in the intermolecular distance near the interface owing to the adsorption of the two surfactant types.

In previous experiments [34], the cationic surfactants with an ester bond (denoted 'Es') and an amide bond (denoted 'Am') have shown different effects, with Es demonstrating a faster propulsive force than Am. Following equilibration, the MD simulations suggested a relatively regular interface with Am compared with Es, as shown in Fig. 4a and b. In order to more quantitatively investigate the differences in the structure of the interface due to the surfactant, the radial distribution function $g(r)$ was calculated using these equilibrium structures (Fig. 4c). These calculations were performed for the centre of mass position of the surfactant molecule and the laurionitrile oil molecule. The results suggest that Am is more tightly packed than Es.

Coarse-grained molecular simulations were conducted based on MD simulations to explore the impact of the surfactant type on oil droplet self-propulsion. These simulations modelled the effect of the surfactant type as a chemical interaction between the oil particles near the interface. The repulsive force a_{ij} between these particles was adjusted, with stronger values applied to the Es model ($a_{oo} = 30$) and weaker values applied to the Am model ($a_{oo} = 22$). These simulations successfully replicated the oil droplet self-propulsion, as depicted by the time variation of the oil droplet's centre of mass (X_{com}) in the x -direction, as shown in Fig. 5. Notably, the Es model exhibited the fastest self-propelled velocity, whereas the Am model demonstrated the slowest, which is consistent with experimental observations. This underscores the influence of the surfactant molecules on the local structure near the interface, which affects the self-propulsion velocity of the oil droplet. The simulation results suggest that using surfactant molecules that enhance the repulsive forces between oil molecules may suppress interfacial Marangoni flow, consequently reducing the droplet's propulsion velocity.

4. Conclusion

The objective of this study was to replicate the self-propelled motion of oil droplets in surfactant systems using DPD simulations, a coarse-grained molecular simulation method. A computational model consisting solely of water and oil particles was used to successfully reproduce the propulsion phenomenon and visualise the internal flow of oil droplets in a theoretical model. Although the actual system and temporal–spatial scales differ, our results demonstrate the feasibility of this approach. Our observations showed that larger droplets had higher propulsion speeds, which is consistent with experimental findings. We also observed self-propelled motion by manipulating the interfacial tension difference at the oil-droplet interface, which Marangoni flow facilitates even with minimal interfacial tension differences.

Additionally, we developed and simulated a model to examine the effects of surfactant molecules on oil droplets. This model is based on the difference in propulsion speeds between two known surfactant systems: cationic surfactants with ester bonds (Es) and amide bonds (Am). The findings indicated that using surfactants such as Es, which cause oil molecules to repel each other and increase the intermolecular

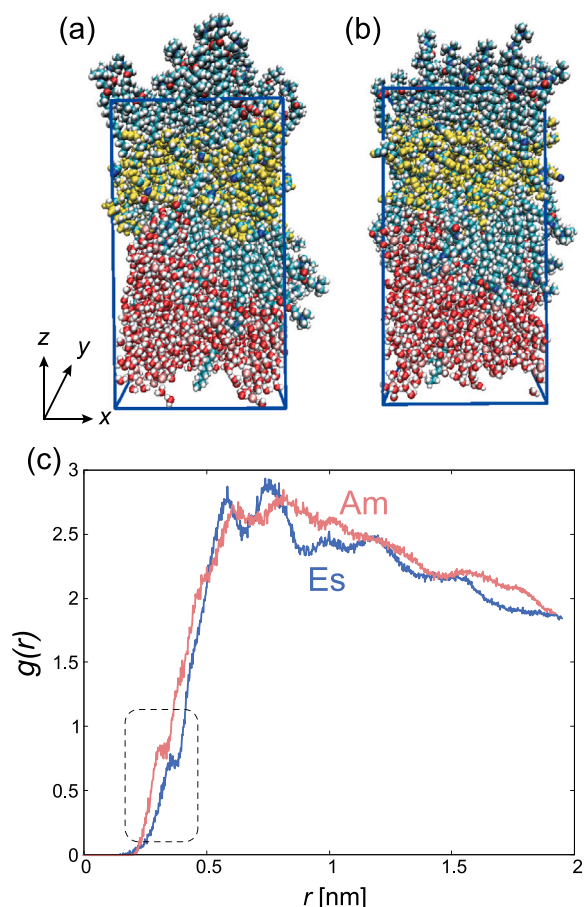


Fig. 4. Snapshots of MD calculations for surfactant–water–oil systems at equilibrium state. We prepared two types of surfactants (indicated by light blue and white beads): (a) an ester bond introduced at the hydrophobic group of the base cationic surfactant denoted 'Es' and (b) an amide bond introduced at the hydrophobic group of the base cationic surfactant denoted 'Am'. The oil molecule employs laurionitrile, indicated by yellow beads; the red and white beads indicate the water molecules. (c) Radial distribution function, $g(r)$, from the centre of mass position between oil and surfactant molecules.

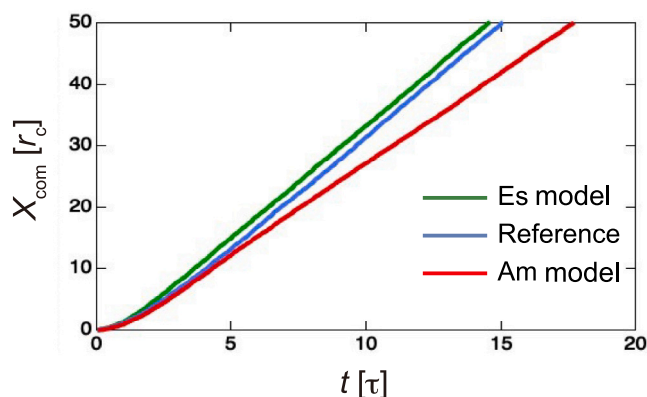


Fig. 5. Dependence of oil particle interaction (a_{oo}) on the time displacement (X_{com}) of the oil droplet centre of mass. The green, blue, and red curves represent strong, reference, and weak repulsive forces, respectively.

distance, resulted in higher propulsion speeds. This section discusses the impact of surfactant molecules on the local structure near the interface, which affects the self-propulsion velocity of oil droplets. These findings emphasise the crucial role of surfactant molecules in the self-propelling mechanism of oil droplets.

CRediT authorship contribution statement

Ken Sasaki: Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Yuuki Ishiwatari:** Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation. **Kazuki Ueno:** Writing – review & editing, Investigation. **Tomoya Kojima:** Writing – review & editing, Validation. **Taisuke Banno:** Writing – review & editing, Validation, Funding acquisition, Conceptualization. **Noriyoshi Arai:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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